Krish Dhage

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972-987-7488
US Citizen

Education

University of Texas at Dallas — Expected: May 2027
Bachelor of Science in Electrical Engineering — ***GPA: 3.70***

- **Relevant Coursework:** Analog Integrated Circuits, Digital Circuits, Electric Network Analysis, Electromagnetic Engineering, Embedded Systems, IoT Lab, RF Circuit Design Principles, Senior Capstone, Signals and Systems, VLSI Design
- **Organizations:** Dallas Formula Racing, Institute of Electrical and Electronics Engineers (IEEE)

Experience

RF Engineering Co-op | ***Mobile Communications America*** — January 2026 – August 2026

- Designed 30+ public safety (ERRCS) and cellular DAS systems in iBwave and AutoCAD across stadiums, data centers, hospitals, and campuses; optimized RF propagation maps and power budgets across antenna distributions to ensure code-compliant and carrier-accepted coverage
- Engineered active fiber-distributed DAS systems and configured complex headends for multi-carrier cellular deployments (AT&T, T-Mobile, Verizon), selecting vendor platforms such as JMA Wireless and NG Comba Flex and calculating signal levels through each passive and active component in the distribution chain
- Interfaced directly with clients and building owners across 5+ projects to present coverage designs, communicate technical requirements, and provide professional recommendations, building hands-on experience in client relations within a professional RF engineering environment

Electrical Team Member | ***Dallas Formula Racing EV*** — August 2024 – December 2025

- Independently designed 3 PCBs for the 2026 FSAE production car, including a 5V–12V boost converter, rear taillight LED assembly, and BSPD circuit; boards were reviewed and fabricated by team leads upon design approval
- Applied full PCB design workflow in LTspice, KiCad, and Altium Designer covering schematic capture, part selection, and layout, with monthly design reviews presented to cross-functional subteams
- Assisted in vehicle wire harness design in SolidWorks Electrical, reviewing connector layouts and routing to support integration of PCB subsystems into the EV electrical architecture

Projects

Half-Wave Dipole Antenna Design | ***RF & EM Simulation*** — June 2026 – August 2026

- Designed a half-wave dipole antenna resonant at 2.4 GHz in Ansys HFSS, targeting 50Ω feed impedance and S11 below −10 dB across the ISM band; set up geometry, material properties, radiation boundaries, and adaptive mesh with frequency sweeps from 2.0–3.0 GHz
- Evaluated return loss, gain, and radiation pattern from simulation results, relating antenna behavior to real-world RF system design principles applied in DAS and wireless coverage engineering

EV Battery Management System | ***BMS Simulation*** — October 2025 – December 2025

- Developed a Python-based BMS simulator in NumPy, SciPy, and Matplotlib modeling SOC estimation, SOH degradation, thermal dynamics, and multi-cell balancing to understand lithium-ion battery operation and lifecycle behavior
- Implemented Coulomb counting and a Kalman Filter for SOC prediction, integrated a temperature-dependent battery model, and structured the simulator as a modular framework for scalable pack analysis

Computer Boot Chip | ***VLSI Design*** — January 2025 – May 2025

- Implemented a finite state machine (FSM) for the computer boot process in Verilog, applying digital design principles to model a real-world boot sequence
- Developed a custom CMOS cell library (inverter, NAND, NOR, AOI/OAI, D flip-flop) in Cadence Virtuoso, mapped and synthesized the FSM onto library cells, verifying timing and logic through waveform simulations

Leadership

Academic Tutor | ***Institute of Electrical and Electronics Engineers – UTD*** — January 2024 – Present

- Tutored 5+ underclassmen in Digital Circuits and Electric Network Analysis, helping students improve exam performance and pass core engineering courses
- Networked with industry professionals and guest speakers at IEEE events and technical workshops, gaining exposure to real-world engineering careers across the DFW engineering community

Technical Skills

Programming & Scripting: Assembly, C/C++, Linux, MATLAB, Python, Verilog, VHDL
Design Tools: Altium Designer, AutoCAD, Cadence OrCAD, iBwave Design, KiCad, SolidWorks Electrical, Virtuoso
Circuit Analysis: HSPICE, LTspice, Multisim, PSpice
RF Networks: DAS/BDA systems, ERRCS, link budget, LTE/5G, RF modeling, signal degradation analysis
Embedded Systems & Protocols: Arduino, ESP32, I^2C, SPI, UART
Laboratory & Test Equipment: AD2, LCR Meter, Logic Analyzer, Multimeter, Oscilloscope, Spectrum Analyzer
Certifications: Comba BDA Certified, FCC GROL, OSHA 10-Hour General Industry Safety